

- 1 (a) By reference to the direction of propagation of energy, state what is meant by a transverse wave.

vibrations/oscillations (of the particles/wave) are perpendicular to the direction (of the propagation of energy)	B1
	.. [1]

- (b) A space telescope is designed to detect electromagnetic radiation with wavelengths in the range $12\mu\text{m}$ to $28\mu\text{m}$.

State the region of the electromagnetic spectrum for this radiation.

Infrared	B1 Allow infra-red, IR Allow slight misspellings e.g. 'infarred' [1]
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- (c) A detector on another space telescope detects an electromagnetic wave. The signal from the detector is transmitted to Earth and displayed on an oscilloscope as shown in Fig. 5.1. The frequency of the signal displayed on the oscilloscope is equal to the frequency of the detected electromagnetic wave.

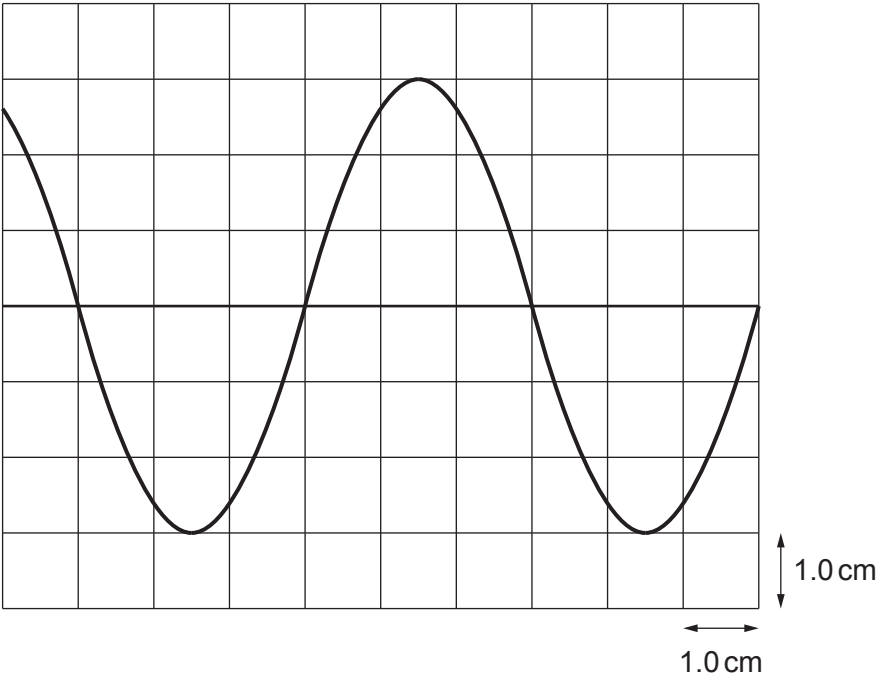


Fig. 5.1

The time-base setting on the oscilloscope is $5.0 \times 10^{-15}\text{s cm}^{-1}$.

Calculate the wavelength of the detected electromagnetic wave.

$T = 6 \times 5.0 \times 10^{-15}$ $T = 3.0 \times 10^{-14}$ $\lambda = cT$ or $\lambda = c/f$ and $f = 1/T$ $\lambda = 3.0 \times 10^8 \times 3.0 \times 10^{-14}$ or $\lambda = 3.0 \times 10^8 / 3.33 \times 10^{13}$ $= 9.0 \times 10^{-6}\text{ m}$	C1 treat missing 10^{-15} as TE and Allow ECF afterwards Allow 't' for 'T' treat all cm to m conversion as a single POT error C1 Any subject Allow v for c Incorrect values of T must be identified as T to imply $f = 1/T$ The C1 marks may be scored independently of one another A1 Allow (bald) 1 SF as answer is exactly $9 \times 10^{-6}\text{ m}$
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wavelength = m [3]

[Total: 5]

2 A progressive transverse wave travelling from left to right is shown at an instant in time in Fig. 4.1.

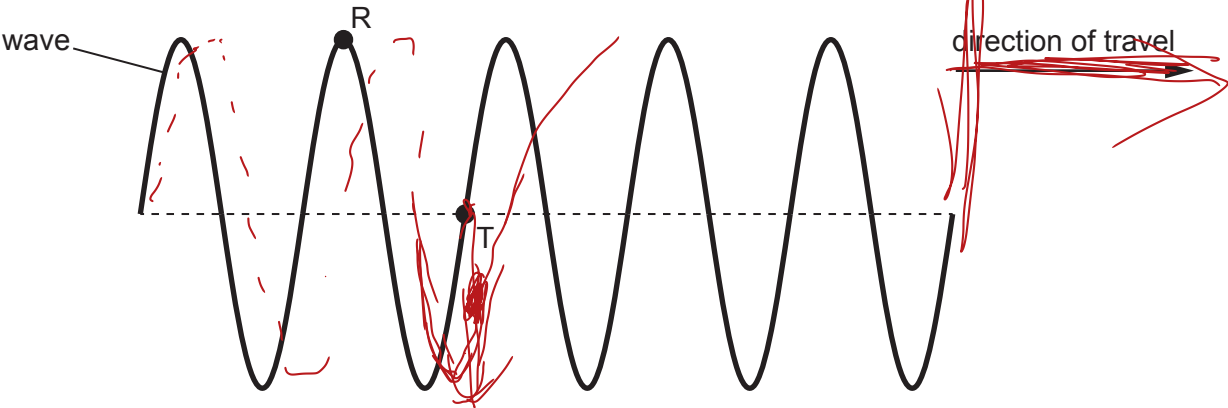


Fig. 4.1

R and T are points on the wave.

(a) State the phase difference between the points R and T.

270°	A1	° [1]
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(b) On Fig. 4.1, draw an arrow at point T to show the direction of movement of point T at the instant shown

arrow pointing vertically downwards at T	A1	[1]
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(c) The horizontal distance between R and T is 0.62 cm, as shown in Fig. 4.2.

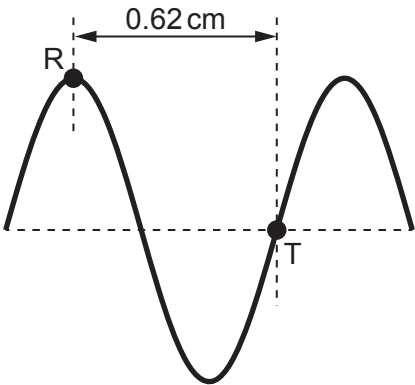


Fig. 4.2 (not to scale)

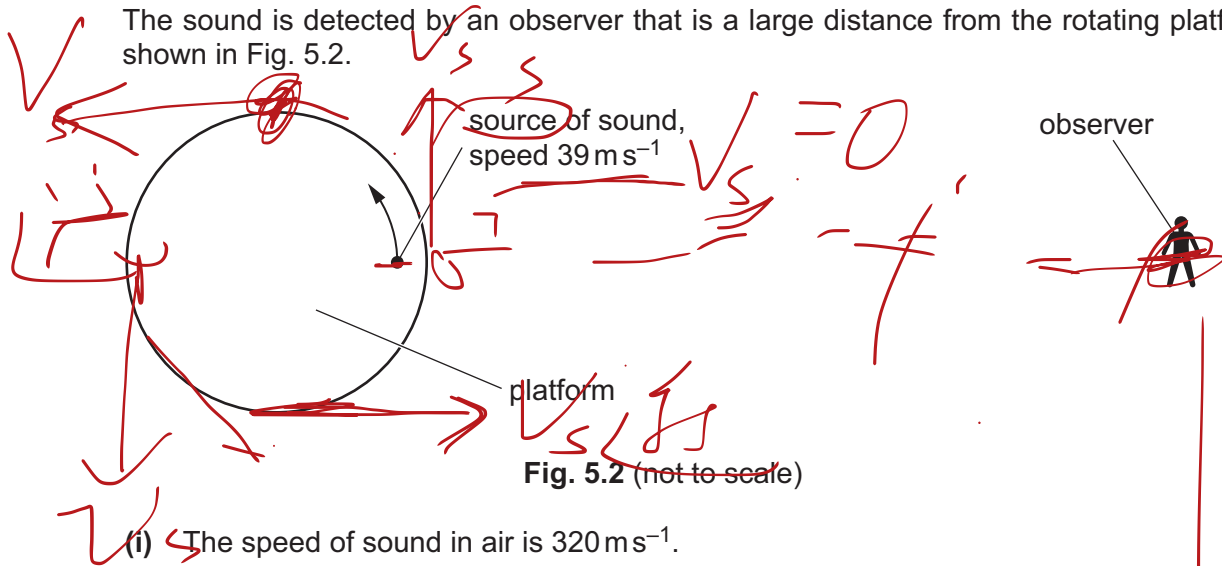
The speed of the wave is 0.27 ms⁻¹.

Calculate the frequency of the wave.

$v = f\lambda$ or $v = \lambda / T$ and $f = 1 / T$	C1
wavelength = $0.62 \times 10^{-2} \times (4 / 3)$ (= 0.83×10^{-2} m)	C1
$f = 0.27 / (0.83 \times 10^{-2})$ = 33 Hz	A1

- (d) A source of sound waves of frequency 780 Hz is on a rotating platform. The speed of the source is 39 m s^{-1} .

The sound is detected by an observer that is a large distance from the rotating platform, as shown in Fig. 5.2.



Calculate the maximum frequency of the sound detected by the observer.

5(b)(i)	$f_o = f_s v / (v - v_s)$	C1
	$f_o = (780 \times 320) / (320 - 39)$	
	maximum frequency = 890 Hz	A1

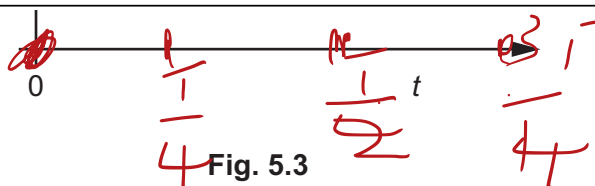
maximum frequency = Hz [2]

- (ii) At time $t = 0$, the observer detects the sound emitted by the source when it was in the position shown in Fig. 5.2.

On Fig. 5.3, sketch the variation with t of the frequency f of the sound detected by the observer for one complete rotation of the platform. Calculations are not required.



5(b)(ii)	line showing f varying both above and below a mean frequency and returning to the original start value of f	B1
	a single cycle of a smoothly oscillating curve of correct phase (starting at mean position, falling to a trough, then rising to a peak, ending at mean position)	B1



[2]

[Total: 8]

- 3 (a) Coherent visible light of a single frequency is incident normally on a double slit. This produces a pattern of bright and dark interference fringes on a screen, as illustrated in Fig. 6.1.

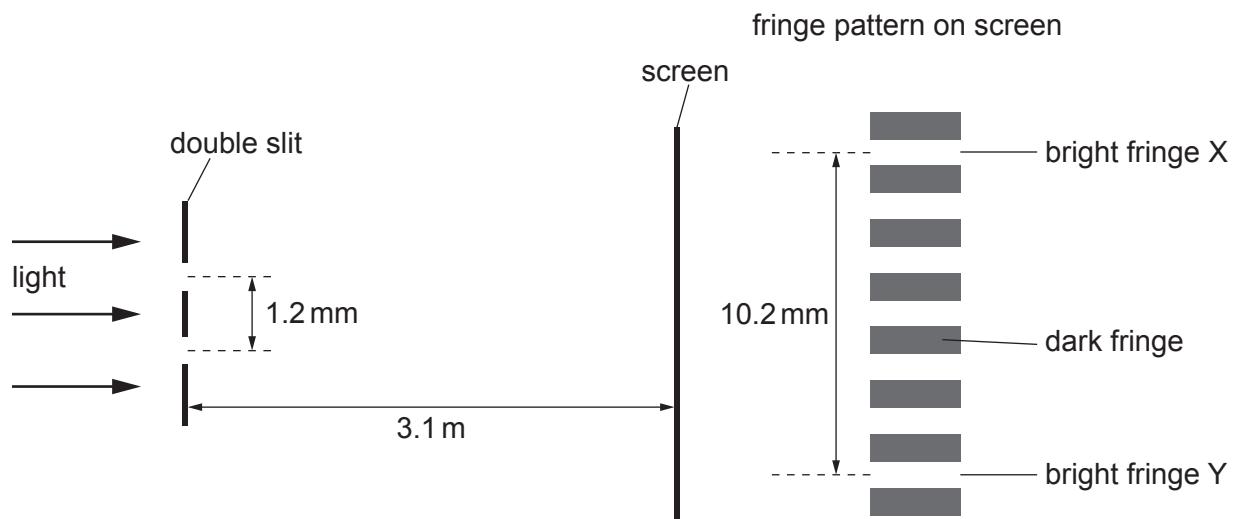


Fig. 6.1 (not to scale)

There are seven bright fringes.

- (i) Explain how the pattern of bright and dark interference fringes is formed.

Any three from:	
• Light diffracts at the (two) slits	
• Light (from each slit) meets / superposes (at the screen)	
• When the phase difference is 0 (degrees) a bright fringe/(intensity) maximum is formed.	
• When the phase difference is 180 (degrees) a dark fringe/(intensity) minimum is formed.	
	B3 [3]

- (ii) The distance between the centres of bright fringe X and bright fringe Y in the pattern is 10.2 mm. The slit spacing is 1.2 mm. The distance from the slits to the screen is 3.1 m.

Calculate the wavelength of the light incident on the slits.

$\lambda = ax/D$ $\lambda = 1.2 \times 10^{-3} \times (10.2 \times 10^{-3} / 6) / 3.1$ $\lambda = 6.6 \times 10^{-7} \text{ m}$	C1 Allow any subject C1 Allow 'y' or 'w' for 'x', allow 'd' for 'D', allow 's' for 'a' A1 Use of $x = 10.2 \times 10^{-3}$ is XP so only first mark can be scored (gives $3.95 \times 10^{-6} \text{ m}$) treat omission of $\times 10^{-3}$ in both a and x as a single POT error <u>Special case:</u> use of $x = 10.2 \times 10^{-3}/7$ to give $\lambda = 5.6 \times 10^{-7} \text{ m}$ scores 2/3 use of $x = 10.2 \times 10^{-3}/5$ to give $\lambda = 7.9 \times 10^{-7} \text{ m}$ scores 2/3	[3]
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- (iii) The light is replaced by different visible light with a shorter wavelength.

State how the new fringe separation will compare to the original fringe separation.

(new fringe separation will be) smaller

B1

..... [1]

(b) The wave is a water wave produced by a dipper S_1 attached to a vibrator in a ripple tank.

An identical dipper S_2 is attached to the same vibrator. The two dippers produce an interference pattern on the water in the tank, as shown in Fig. 4.3.

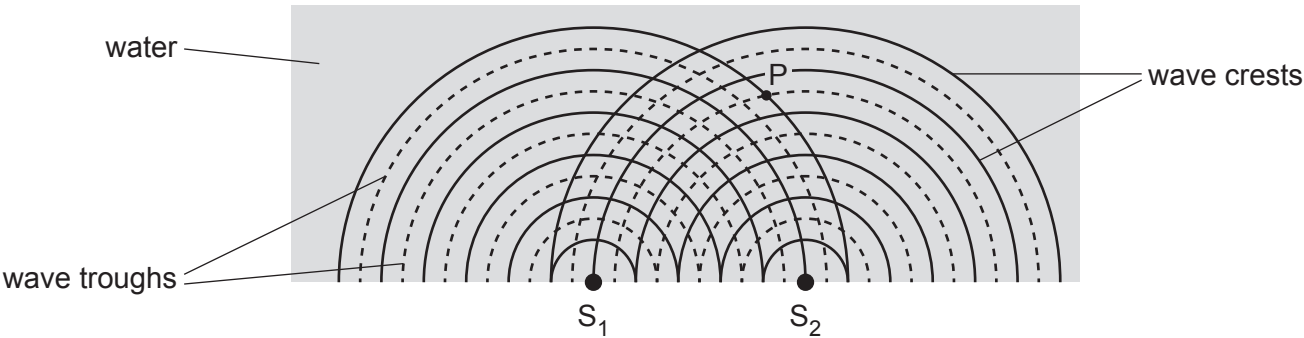


Fig. 4.3 (not to scale)

The wave crests from each source are represented by solid lines on Fig. 4.3 and the wave troughs are represented by dashed lines.

At point P in Fig. 4.3, the wave from S_1 has the same amplitude A as the wave from S_2 .

Describe and explain the amplitude of the resultant wave at point P.

resultant displacement is the sum of the displacements of the waves (from S_1 and S_2) or waves (from S_1 and S_2) <u>superpose</u> (at P)	B1
Any one point from: • (at P) the waves (from the two sources) (always) destructively interfere • (at P) the waves have a path difference that is (always) an odd number of half-wavelengths / their path difference is one and a half wavelengths • (at P) the waves have a phase difference that is (always) 180° / they are in antiphase / crest of one wave meets trough of other wave	B1 [3]
amplitude (of the resultant wave) is zero (at all times)	B1 : 8]

- 4 Two coherent sources X and Y of microwaves of frequency 2.5×10^{10} Hz are a distance of 0.18 m apart in a vacuum, as shown in Fig. 6.1.

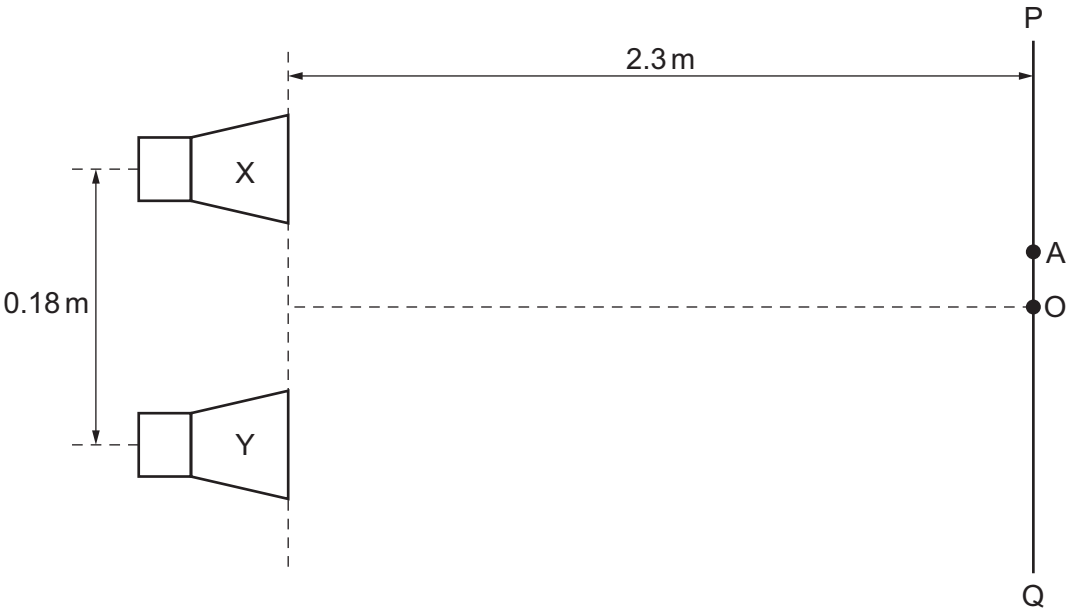


Fig. 6.1 (not to scale)

There is a phase difference of 90° between the waves emitted at the two sources.

A microwave detector moves along the line PQ, which is parallel to the line joining the two sources and 2.3 m away from it.

Point O is on the line PQ at a position that is equidistant from the two sources.
Point A is the position on line PQ where the intensity of the microwaves is the greatest.

- (a) (i) Explain why the position of greatest intensity is **not** at point O.

for maximum intensity, waves must be in phase (at detection)	B1
(phase difference at source means) waves are not in phase / have a 90° phase difference <u>at O</u>	B1

..... [2]

- (ii) On Fig. 6.1, draw a cross (x) to show the position of the point on line PQ where the intensity minimum that is the closest to point O occurs. Label this point B. [2]

point B labelled on the line PQ below O	M1
distance OB = distance OA	A1

(b) (i) Show that the wavelength of the microwaves is 0.012 m.

$v = f\lambda$	C1
$\lambda = (3.00 \times 10^8) / (2.5 \times 10^{10}) = 0.012 \text{ m}$	A1

[2]

(ii) For point A on line PQ, determine the difference in the distances Δx travelled by the microwaves from X and the microwaves from Y.

$\Delta x = 0.012 / 4$ $= 0.0030 \text{ m}$	A1
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$\Delta x = \dots\dots\dots \text{m}$ [1]

(iii) Use the formula for the double-slit interference of light to calculate the distance between adjacent intensity maxima on line PQ.

$\lambda = ax / D$	C1
distance = $\lambda D / a = 0.012 \times 2.3 / 0.18$ $= 0.15 \text{ m}$	A1

distance = $\dots\dots\dots \text{m}$ [2]

[Total: 9]

- 5 Light of a single frequency is incident normally on a diffraction grating. An interference pattern of bright and dark fringes forms on the semicircular screen shown in Fig. 6.1.

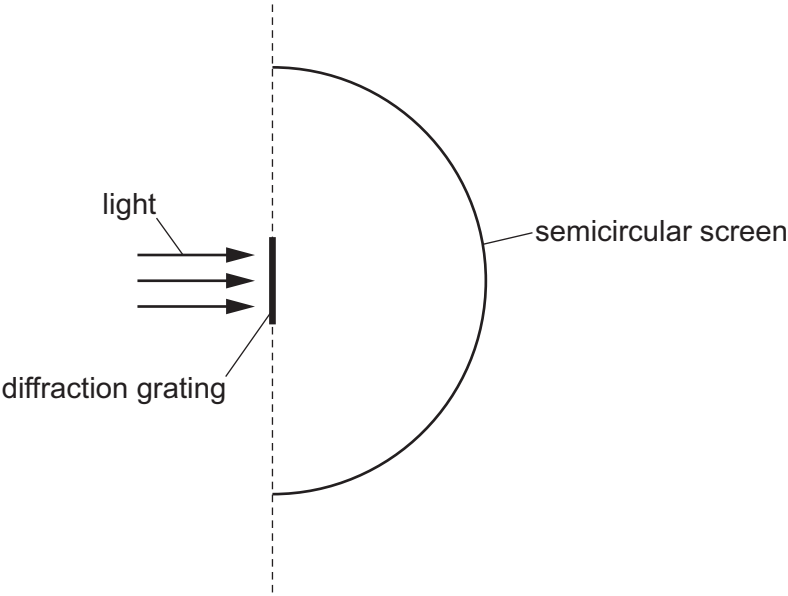


Fig. 6.1 (not to scale)

The light has wavelength 520 nm.
The separation of the lines in the grating is 3.8×10^{-6} m.

- (a) Determine the total number of bright fringes formed on the screen.

$n\lambda = d \sin \theta$	C1	3]
$n = (3.8 \times 10^{-6} \times \sin 90^\circ) / (520 \times 10^{-9})$ (= 7.3)	C1	
number of bright fringes formed = 15 (given as an integer)	A1	

- (b) The light is replaced with red light of a single frequency.

- (i) State whether the frequency of the red light is greater than, less than or the same as the frequency of the original light.

(frequency of red light is) less (than frequency of original light)	B1	[1]
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- (ii) State and explain the effect of this change on the number of bright fringes formed on the screen. A calculation is not required.

(red light has) longer wavelength	M1
(so) number (of bright fringes formed) is less / fewer (bright fringes are formed)	A1

.....
..... [2]

[Total: 6]

- 6 (a) A stationary wave is formed on a string XY that has a length of 0.48 m. Fig. 5.1 shows the string at one instant in time.

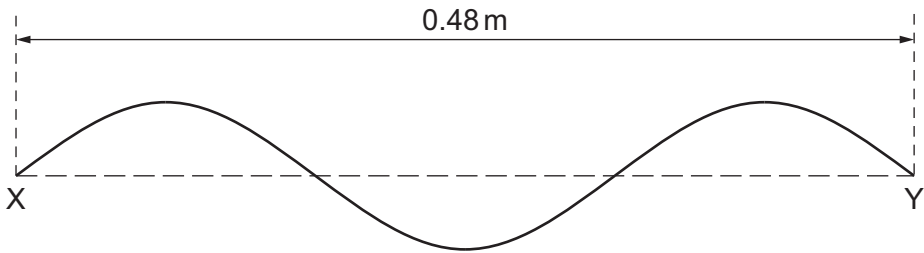


Fig. 5.1

The speed of the wave on the string is 1400 ms^{-1} .

- (i) On Fig. 5.1, draw a cross (×) at **one** position that is a node and another cross at **one** position that is an antinode. Label the node N and the antinode A. [1]

- (ii) Show that the wavelength of the wave produced is 0.32 m. Explain your reasoning.

5(a)(i)	cross labelled N marked at the intersection of the solid and dashed lines or at X or Y and cross labelled A marked at a peak or a trough	B1
5(a)(ii)	(XY is 1.5 wavelengths, so) wavelength = $0.48 \times (2/3) = 0.32 \text{ m}$ or (wavelength is twice node–node distance so) $\lambda/2 = 0.48/3 = 0.16 \text{ m}$ and wavelength = $0.16 \times 2 = 0.32 \text{ m}$	B1

[1]

- (iii) Calculate the frequency of the wave.

5(a)(iii)	$v = f\lambda$	C1
	frequency = $1400 / 0.32$ = 4400 Hz	A1

Hz [2]

- (b) A stationary wave is formed on a stretched string AB, as shown in Fig. 6.2.

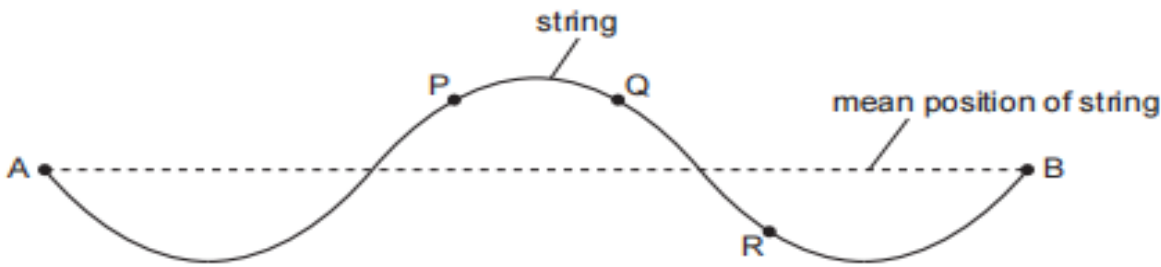


Fig. 6.2

P, Q and R are points on the string.

- (i) On Fig. 6.2, draw a cross (×) to show the position of a node. [1]

- (ii) State the phase difference between P and Q.

phase difference = ° [1]

- (iii) State the phase difference between P and R.

A cross at the intersection of the string and the mean position line	B1	Allow cross at A or B Allow centre of cross to be within 2mm of intersection. Judge by eye If more than one cross drawn, all must be correct	 ° [1]
0	A1	Allow degrees or radians not 360° or 2π radians	[Total: 10]
180°	A1	Allow pi radians (unit must be explicit)	

- 7 A stretched string PQ has length 1.2 m. One end of the string is attached to a vibration generator and the other end is attached to a wall, as shown in Fig. 5.1.

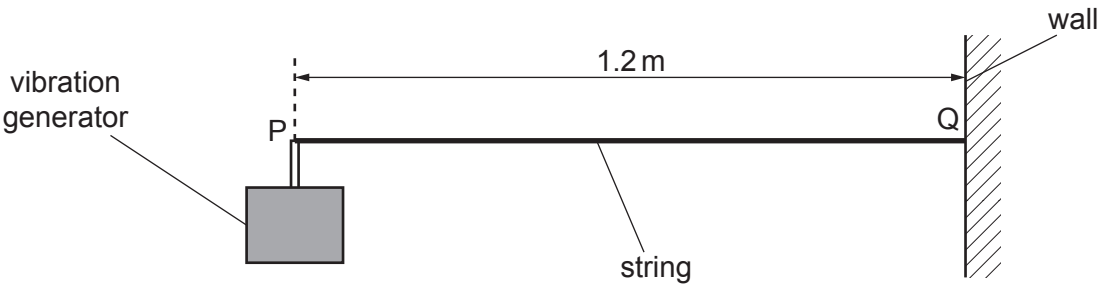


Fig. 5.1

The vibration generator is switched on and a stationary wave is formed on the string. The string is shown at one instant of time in Fig. 5.2.

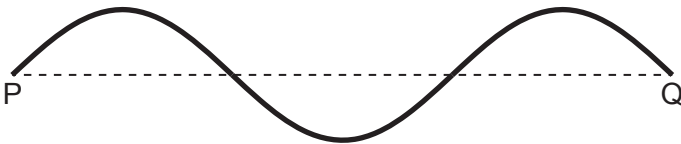


Fig. 5.2

- (a) Explain how a stationary wave is formed between the vibration generator and the wall.

wave(s) (travel along string and) reflect at fixed point / wall / Q / end / vibration generator / P	B1
incident and reflected waves superpose	B1

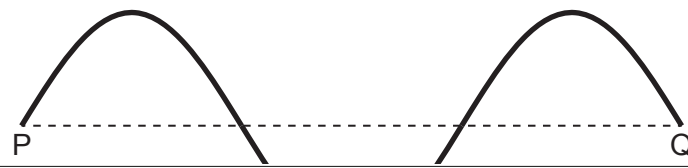
.....
..... [2]

- (b) Calculate the wavelength of the stationary wave shown in Fig. 5.2.

0.80 m	A1
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wavelength = m [1]

- (c) Fig. 5.3 shows the stationary wave at time $t = 0$ when all points on the wave are at their maximum displacements.



same wavelength as original throughout and passing through intersection of solid and dashed lines	B1
reflected in dashed line and of same amplitude	B1

Fig. 5.3

The period of the wave is 0.16 s.

On Fig. 5.3, sketch the shape of the stationary wave at time $t = 0.24$ s.

[2]

- (d) Points R and T on the string are a horizontal distance of 0.30 m apart and in the positions shown in Fig. 5.4.

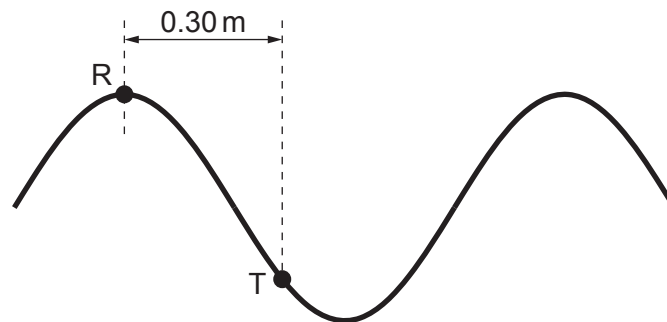


Fig. 5.4

State the phase difference between the oscillations of points R and T.

180°	A1
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phase difference = ° [1]

- (e) Calculate the speed of the progressive waves on the stretched string.

$v = f\lambda$ and $f = 1/T$ or $v = \lambda / T$	C1
$v = 6.25 \times 0.80$ or $0.80 / 0.16$ $= 5.0 \text{ m s}^{-1}$	A1

speed = m s^{-1} [2]

[Total: 8]

- 8 (a) With reference to the direction of transfer of energy, compare the oscillations of transverse and longitudinal progressive waves.

longitudinal waves have oscillations <u>parallel</u> to the (direction of) transfer of energy	B1
transverse waves have oscillations <u>perpendicular</u> to the (direction of) transfer of energy	B1

..... [2]

- (b) A pipe is open at one end and closed at the other with a piston. The piston can slide freely and is at a distance of $4.5 \times 10^{-2}\text{m}$ from the open end of the pipe.

A loudspeaker is positioned near the open end of the pipe and emits a sound wave of a single constant frequency. A stationary wave is formed in the pipe, as illustrated in Fig. 4.1.

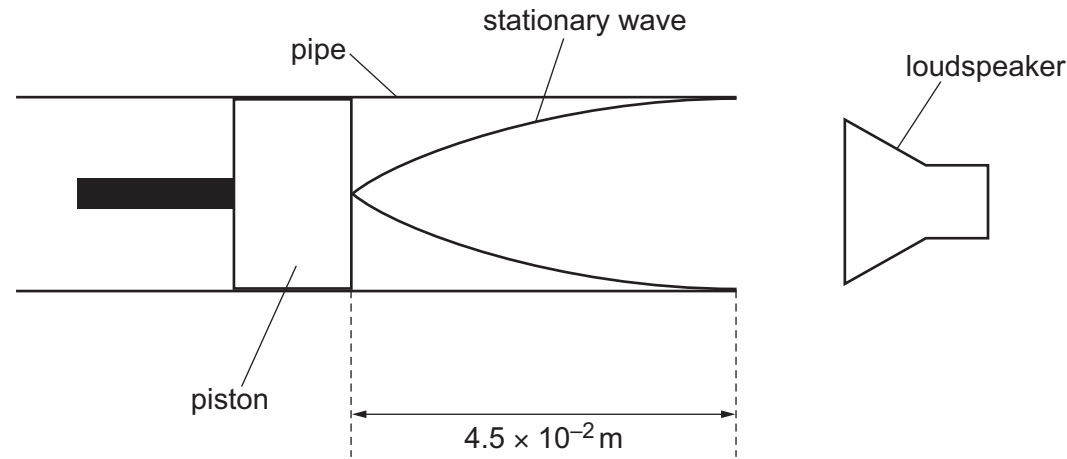


Fig. 4.1

- (i) On Fig. 4.1, draw a letter A at the position of an antinode. [1]
- (ii) The speed of sound in air is 340ms^{-1} .

Determine the frequency of the sound wave.

4(b)(i)	A marked at the open end of the pipe	B1
4(b)(ii)	$f = v / \lambda$	C1
	$\lambda = 4 \times 4.5 \times 10^{-2}$	C1
	$f = 340 / (4 \times 4.5 \times 10^{-2})$	A1
	$= 1900\text{ Hz}$	

- (iii) The piston is moved to the left. The frequency of the sound wave emitted by the loudspeaker is then changed so that a stationary wave is formed with same number of antinodes as in Fig. 4.1. Hz [3]

State and explain the change that is made to the frequency of the sound wave.

the node–antinode distance is longer or the wavelength (of the wave) is longer	M1	
(the speed of sound is constant so) the frequency (of the wave) is lower	A1	

..... [2]

[Total: 8]